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Design and evaluation of Avatar: An ultra-low-latency immersive human–machine interface for teleoperation☆

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ABSTRACT

Spatially separated teleoperation is crucial for inaccessible or hazardous scenarios but requires intuitive human–machine interfaces (HMIs) to ensure situational awareness, especially visual perception. While 360° panoramic vision offers immersion and a wide field of view, its high latency reduces efficiency and quality and causes motion sickness. This paper presents the Avatar system, an ultra-low-latency panoramic vision platform for teleoperation and telepresence. Using a convenient method, Avatar's measured capture-to-display latency is only 220 ms. Two experiments with 43 participants demonstrated that Avatar achieves near-scene perception efficiency in near-field visual search. Its ultra-low latency also ensured high efficiency and quality in teleoperation tasks. Analysis of subjective questionnaires and physiological indicators confirmed that Avatar provides operators with intense immersion and presence. The system's design and verification guide future universal, efficient HMI development for diverse applications.

1. Introduction

Teleoperation enables human operators to control robots or devices from a distance with some form of separation [1]. Spatially separated teleoperation is important when human presence is impossible or dangerous, such as space exploration, deep-sea operations, and hazardous environments like nuclear sites and coal mines. It is also a possible solution for applications that require highly specialized skills but are short of personnel [2].

The rapid development of robotics and AI algorithms in recent years has brought more application possibilities for teleoperation. However, it also continuously poses new requirements for the design of human–machine interfaces (HMI). A key challenge in developing HMI for spatially separated teleoperation is effectively presenting data about the robot and environment to the human operator to ensure their situational awareness [3]. Visualization methods via immersive virtual reality (VR) and augmented reality (AR), especially with head-mounted displays (HMDs), outperform traditional liquid crystal display (LCD) [4] due to providing better three-dimensional (3D) spatial awareness [5] and immersion [6,7]. Using HMDs head tracking to decouple visual direction control from the robot's movements is more intuitive-friendly and efficient than switching visual direction by hand or coupled with the robot's movements in general applications, especially for non-experienced users [8].

However, the use of HMDs in HMI for spatially separated teleoperation face the challenge of "HMD pose latency"[9], which refers to the time delay between the operator's head movement and the corresponding change in the visual perspective. Such latency can lead to a mismatch between the operator's visual perception and physical actions, resulting in simulation sickness [10]. Panoramic images and videos capture visual information from all directions, significantly enhancing the operator's field of view (FOV) and providing greater immersion. Fiala [9] also verified that panoramic vision effectively addresses the issue of "Head-Mounted Display (HMD) pose latency", eliminating delays during perspective shifts of the human operator, which may reduce simulation sickness caused by conflicts between visual and inertial stimuli [11,12], particularly when using high-refresh-rate HMDs. However, in previous studies, panoramic vision systems combined with HMDs suffer from low resolution, poor stitched image and high capture-to-display latency [13–16], which can be attributed to the time-consuming stitching process and the large amount of data

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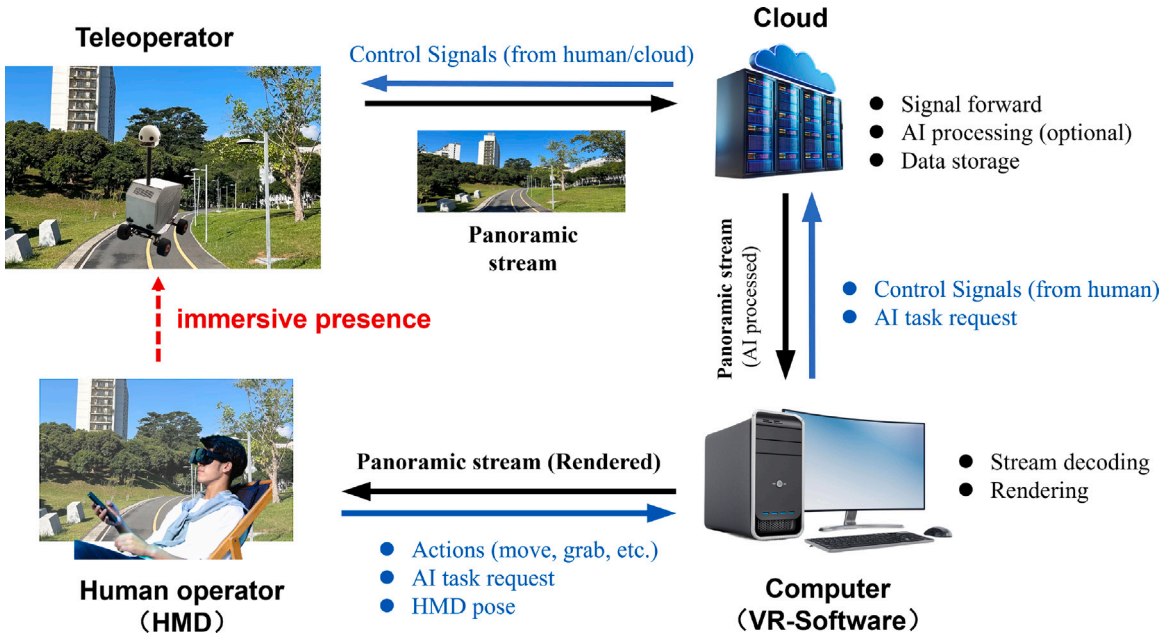


Fig. 1. The framework of the Avatar system.

that needs to be transmitted. The lack of ultra-low capture-to-display latency panoramic vision systems can lead to the failure to make an objective and fair evaluation of the human–machine interface of the panoramic visual solution.

This paper designed and evaluated the Avatar system, whose framework is illustrated in Section 1. This system is based on a panoramic camera rig, 5G network transmission, cloud servers, and virtual reality (VR) technology, resulting in an immersive, ultra-low latency panoramic human–computer interaction system for spatially separated teleoperation. We conducted experiments to evaluate the system's performance in standard tests for response robots and teleoperation tasks. We quantitatively compared the task execution efficiency and quality of participants when using Avatar and different HMIs. At the same time, subjective questionnaires and physiological indicators measurements were introduced to evaluate the sense of presence and immersion provided by HMIs in teleoperation tasks.

The paper is organized as follows. Section 2 summarizes related work on the vision of HMI and latency of teleoperation. Section 3 briefly introduces the system architecture and key methods. Section 4 presents the evaluation apparatus, results, and analysis of tests on uplink latency, standard test for response robots and teleoperation on dynamic scenes. Finally, section. 5 discusses the conclusions, limitations and prospects.

The main contributions of this study are as follows:

1. We developed an immersive HMI by integrating panoramic vision and virtual reality (VR) techniques for spatially separated teleoperation. Our system achieves an ultra-low uplink latency of only 220 ms for LCD and 250 ms for HMD, the lowest for a panoramic vision system reported in the literature, meeting requirements for timely teleoperations across various applications.
2. We quantitatively evaluated the near-field perception efficiency of the Avatar system and two widely adopted HMI methods using standard tests. The experiments show that the visual solution of the Avatar system has the near-field perception efficiency closest to on-site perception. The average time to complete the perception task is only 1.1 times that of being at the scene.
3. We quantitatively evaluated the performance of the Avatar system in the teleoperation task. Meanwhile, by analyzing subjective questionnaires and physiological indicators, we comprehensively evaluated the participants' immersion and sense of presence. The analysis results show that the ultra-low uplink latency of the Avatar system enables the vast majority of subjects to complete the teleoperation task with good efficiency and quality. The HMI using HMD provides better immersion than LCD, making users more focused. However, the HMD solution does not show an advantage in teleoperation efficiency and quality. We analyzed and summarized the existing limitations, guiding further optimizing the HMIs.

2. Related work

2.1. Vision in human–machine interfaces for teleoperation

Designing effective HMI remains challenging for teleoperation systems [17]. Most user interface developers are experts in robotics and human–robot interaction. However, they may lack expertise in data visualization, which involves presenting robot and environment data in a way that is easy to understand for human operators [3]. High-quality vision perception is a hot topic in teleoperation human–machine interface because it directly affects the operator's situational awareness [18]. Traditional methods using 2D display to present visual data from the robot's perspective can cause "keyhole" (also known as "soda straw") effects due to limited viewing angles [19]. Showing multiple cameras simultaneously on a LCD allows users to combine multiple vision messages to avoid this effect, but can raise cognitive overload and mission complexity [20], requiring specialized training for human operators [21].

Virtual reality (VR) technologies can provide 3D interactions and immersive experiences, possibly solving these challenges. The technological development of VR devices has made their prices more affordable, leading to increased research on human–machine interfaces for teleoperation based on VR technology [22]. Multiple studies have demonstrated the advantages of virtual reality compared to traditional interfaces, such as the combination of a keyboard, mouse, and monitor, in terms of task completion time, workload, usability, and preference ratings [23] for most teleoperation tasks.

Most virtual reality interfaces use HMDs as displays, providing a completely immersive environment for human operators [6]. Immersion improves situational awareness and comprehension, helping the operator to focus better on teleoperation tasks [24]. Directly presenting video streaming in HMDs can link human operator with the

robot's perspective, which is considered to be more intuitive and cost-efficient [25–28], also more time-effective than building digital twins or 3D reconstruction in a virtual space [29,30]. Therefore, it is widely used in various interface designs [31].

Streaming for HMDs includes using monocular, stereo, or panoramic cameras. Monocular and stereo cameras are usually set on Pan-Tilt-Unit (PTU), the yaw and pitch angles of the cameras are synchronized by capturing the head motion by HMDs [32–34], which has been proved to be more intuitive and friendly for non-experienced user in general tasks than using joystick or keyboard to switch visual direction in a decoupled vision-motion relationship [8]. Ikei [35] set two panoramic cameras on a pan-unit to generate panoramic stereo vision. However, such solutions face the problem of "HMD pose latency". This latency is equal to the end-to-end latency of the teleoperation system [9]. In spatially separated teleoperation applications, especially for mobile robots using non-cable networks, the end-to-end latencies can easily exceed the human's tolerance threshold (about 17 ms) [11] and cause simulation sickness [10]. Streaming a panoramic video in HMDs can effectively address this issue [9]. Since the computing power of modern computers is sufficient to complete HMD rendering in an extremely short time, the perspective switching time is mainly limited by the refresh rate of the HMDs, for most HMDs like VIVE PRO2, Pico 4, and Apple VisionPro, the refresh rates are higher than 60 Hz, where the perspective switch rendering can be finished within 17 ms and avoid simulation sickness.

There is a rich literature on panoramic vision systems' research on mobile telepresence or teleoperation robots. Zhang [36] and Hansen [37] proposed a 360° panoramic vision telepresence system with an HMD. Teo [38,39] proposed a system combining 360° panoramic vision and 3D reconstruction for telecollaboration. Experimental results demonstrate that users perform better in 360° panoramic vision. 360° panoramic vision reported the widest viewing angle, least vertigo, and smoothest perspective switch in Luo's evaluation [13] compared to monocular and stereo, but suffers from high latency and insufficient resolution. Real-time transmission of 360° video brings computational burden and bandwidth challenges [40,41], which may lead to high latency to prevent such systems from application in teleoperations. However, only a few research mention or focus on such issues, which will be discussed in section. 2.2. Compared to most existing 360° panoramic mobile telepresence systems, this paper focuses on establishing and evaluating an ultra-low latency panoramic vision system "Avatar" for telepresence and teleoperation.

2.2. Latency of teleoperation system

Latency arises from the cumulative delays introduced by each component of the teleoperation system. Each stage accumulates to the end-to-end latency, from input capture and processing to hardware and transmission. According to Kamtam [42], the end-to-end latency of a teleoperation system consists of two components: uplink and downlink. Uplink latency refers to the total delays experienced from the moment that the camera captures an image until it is displayed, often referred to as capture-to-display, glass-to-glass, or event-to-eye latency. Downlink latency encompasses the total delays from transmitting operator control commands to the corresponding robotic response, commonly known as control-to-response latency.

Blackett [43] verified that the tolerance threshold and impact of latency could vary significantly across different application scenarios and missions. For instance, latencies of 1.5 to 3 s may similarly impact tele-flying, while a latency of 170 ms can affect a tele-driving mission. Introducing a 3-second latency in a box extraction task resulted in a 60% increase in the task completion time compared to scenarios without delay. In contrast, a similar latency in a "peg in hole" positioning task led to a 213% increase in completion time [44]. Kamtam [42] also noted that both the stability and magnitude of end-to-end latency impact the efficiency and quality of teleoperations and telepresence.

Since most teleoperation applications involve tele-driving, interaction, and manipulation, latency must be sufficiently low to support effective teleoperation. A latency of just 300 ms can lead to noticeable changes in human operator behavior, while a latency of 1 s may cause operators to shift from a continuous command strategy to a "move and wait" approach [45]. Lum [46] conducted experiments on varying latency levels for surgical robotics in teleoperation across America, Europe, and Japan. The results indicated that as latency increases from 0 to 0.5 s, both completion time and the length of the tool tip trajectory increase. Specifically, a latency of 0.25 s results in a 1.45-fold increase in completion time and a 1.28-fold increase in trajectory length. When latency reaches 0.5 s, these factors rise to 2.04 for time and 1.53 for trajectory length. Neumeier [47] researched the impact of latency in automated driving, demonstrating that latencies exceeding 300 ms significantly negatively affect tele-driving.

Furthermore, fluctuations in latency diminish the ability to teleoperate the vehicle effectively. Zhang [48] noted that in a tele-driving scenario, steady latency below 170 ms is manageable, while latency not exceeding 300 ms is also handleable, consistent with Neumeier's findings [47]. The latency between 300 and 500 ms poses challenges for tele-driving, even at low speeds, while latencies over 700 ms render the task nearly impossible. Based on these findings, general teleoperation systems should aim for a steady end-to-end latency of less than 300 ms, ideally targeting 170 ms or lower.

However, only a few previous studies on teleoperation systems have explicitly analyzed the system's latency [31]. For mobile teleoperation systems based on wireless network connection, TELEOPERATIVO [49] reached an average uplink latency of 180 ms with a stereo camera, which is the lowest uplink latency reported in the literature. Other researchers reported 350 ms [50] and 800 ms [51] on a monocular camera. For a panoramic visual system, Zheng [14] developed a live panoramic immersive visual system utilizing HTC VIVE VR equipment and an eight-lens camera, resulting in an average capture-to-display latency of 2.9 s. Du [15] achieved latencies ranging from 1.5 to 3 s for their immersive teleoperation system. Taleb [16] reached an uplink latency of 600 ms, which is the lowest among panoramic vision systems but still unacceptably high for most teleoperation tasks.

Our Avatar system achieves an uplink latency of only 220 ms, the lowest for a panoramic vision system reported in the literature, meeting requirements for timely teleoperations across various applications.

3. System overview

There are two key factors in achieving the performance of the Avatar system. First, the realization of ultra-low uplink latency ensures that the end-to-end latency of the teleoperation system does not exceed 300 ms; this involves both algorithm design and hardware implementation throughout the entire uplink process. Second, the system design adopts a panoramic camera as the video stream hardware for the HMD interface, which effectively addresses the issue of HMD pose latency and thereby enhances user immersion. The combination of ultra-low uplink latency also effectively reduces user motion sickness. Although panoramic cameras and VR HMDs are both mature products, the Avatar system has optimized both algorithms and hardware for panoramic streaming latency, achieving a state-of-the-art latency level. As a result, when used with HMDs, the system demonstrates excellent human–machine interaction performance. In this section, we will focus on describing the construction methods and key aspects of the system, while retaining the necessary introduction to the system architecture.

The Avatar system consists of three components: the teleoperators (mobile robotics), the cloud, and the human operator with HMIs (computers equipped with VR devices). The structure is illustrated in Fig. 1.

The system's uplink processing workflow includes the following steps: image acquisition, panoramic stitching, video encoding, network transmission, cloud forwarding, network transmission, and software

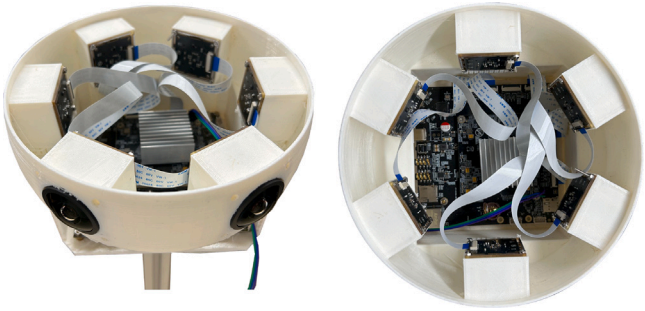


Fig. 2. The panoramic camera rig and edge computing device of Avatar system.

streaming decoding for display. The real-time panoramic stitching algorithm will be described in detail in section. 3.1, and the video coding and decoding algorithm utilizes the High-Efficiency Video Coding (HEVC) standard [52]. The network transmission from the teleoperator to the cloud services is facilitated by Customer Premise Equipment (CPE), with tests showing that our upload bandwidth for 4K panoramic video is approximately 6Mbps. The sufficient bandwidth provided by the CPE ensures the timely upload of high-definition panoramic video, thereby guaranteeing ultra-low uplink latency. Real-time rendering on the computer is facilitated by a high-performance computer equipped with GPU and VR hardware. We will describe these components in detail in section. 3.2.

3.1. Vision of the Avatar system

Panoramic streaming needs efficient hardware and real-time processing algorithms. The vision system of the Avatar is based on a camera rig consisting of six lenses designed to create panoramic live video. The camera rig is illustrated in Fig. 2.

Traditional USB cameras encode the images after capture. When the edge processing device retrieves the images via the UVC protocol, it must decode and convert them to YUV format before proceeding with subsequent processing steps. To minimize latency, our cameras employ a Mobile Industry Processor Interface (MIPI) to directly connect the CMOS sensor to the edge computing device, as shown in Fig. 2. The integrated Image Signal Processor (ISP) module on the edge computing device performs algorithmic processing on the raw images obtained from photoelectric conversion, including Black Level Correction, Lens Shading Correction, Bad Pixel Correction, Demosaicing, Auto White Balance, Color Correction and Gamma Correction [53]. The MIPI method eliminates the latency associated with additional encoding, decoding, and memory copying with USB protocol camera modules. Six frames are time-aligned and stitched after the ISP processes the YUV data.

To reach real-time performance, we adopt the LUT (Look-Up Table) stitching algorithm [54,55] to align camera video streams. Alignment steps can be executed offline without introducing latency to the online stitching process.

The stitching process can be described as follows. Let the kth frame image from camera i be denoted as $\boldsymbol{C}_i(k)$, the kth frame of the stitched panorama for display as $\boldsymbol{P}(k)$, the warped frame of $\boldsymbol{C}_i(k)$ as $\boldsymbol{P}_i(k)$, and the blending mask for $\boldsymbol{P}_i(k)$ as $\boldsymbol{M}_i(k)$. $\boldsymbol{P}_i(k)$, $\boldsymbol{M}_i(k)$ and $\boldsymbol{P}(k)$ share the same coordinates in a 2D space. To generate $\boldsymbol{P}(k)$ from $\boldsymbol{P}_i(k)$, we employ a linear blending method that computes $\boldsymbol{M}_i(k)$ to handle the overlapping areas between $\boldsymbol{P}_i(k)$, $\boldsymbol{P}_{i-1}(k)$ and $\boldsymbol{P}_{i+1}(k)$, thus producing a seamless panoramic frame.

The first step involves obtaining $\boldsymbol{C}_i(k)$ from the cameras. We utilize six fisheye lenses to achieve a larger field of view (FOV). Although fisheye lenses can introduce significant distortion in the peripheral areas during imaging, existing fisheye camera models and image distortion correction techniques can partially mitigate this distortion. The images are then transformed into equirectangular images $\boldsymbol{C}_i(k)$. The images obtained from the six lenses are displayed in Fig. 3.

The second step is to establish panoramic coordinates for $\boldsymbol{P}_i(k)$, $\boldsymbol{M}_i(k)$ and $\boldsymbol{P}(k)$, and to compute $\boldsymbol{P}_i(k)$ from $\boldsymbol{C}_i(k)$. Inspired by Saeed [54,55], we use a LUTs-based method to store the pixel mapping relationship between $\boldsymbol{C}_i(k)$ and $\boldsymbol{P}_i(k)$, which can be expressed as a transformation $\boldsymbol{P}_i(k) = \mathscr{T}\ [\boldsymbol{C}_i(k)]$. This process begins with extracting feature points from $\boldsymbol{C}_i(k)$ and matching them to find correlations between $\boldsymbol{C}_{i-1}(k)$, $\boldsymbol{C}_i(k)$ and $\boldsymbol{C}_{i+1}(k)$ using the K-Nearest Neighbors (KNN) algorithm. Random Sample Consensus (RANSAC) is employed to estimate the correlation matrix, commonly called homography estimation. Finally, Levenberg–Marquardt optimization is used to determine the best LUTs for our camera rig, starting with the initial value of the homography matrix estimated earlier.

The third step involves computing the blending mask $\boldsymbol{M}_i(k)$. The linear blending method processes the masks of two images in the overlapping region to achieve a smooth transition, thereby reducing visual defects. The overlapping areas between each mask and the preceding and following masks are processed cyclically from 1 to 6, resulting in the corresponding blending masks $\boldsymbol{M}_1(k)$ to $\boldsymbol{M}_6(k)$ for each image, contributing to the overall panoramic image.

Following the steps above, a panoramic view obtained online is shown in Fig. 3.

3.2. Components of the Avatar system

3.2.1. Teleoperators

Teleoperators enable remote execution of tasks in inaccessible locations. The Avatar system exemplifies this through a mobile robotic platform that integrates a panoramic vision system. As depicted in Fig. 4, our implementation comprises five core components: (1) Panoramic camera rig; (2) An edge-computing unit handling stream processing; (3) A 5G network module capable handling 100Mbps bandwidth; (4) A mobile platform capable to carry hardwares on teleoperator; and (5) Actuators including 6-DoF robotic arm coupled with dexterous hands.

3.2.2. Cloud services

The cloud platform exchange data between teleoperators and human operators over the Internet or private networks. It provides essential services, including data storage, processing, and execution of AI algorithms. Within the Avatar framework, cloud capabilities are fully customizable and scalable. Users dynamically provision GPU resources, deploy tailored AI modules for specialized computations and establish dedicated data archives/backups based on application-specific demands. While AI-enhanced panoramic vision offers significant benefits [56], it must be noted that associated processing steps (decoding, AI inference, re-encoding) contribute to uplink latency. We deployed a private cloud server with a 112-core Intel (R) Xeon (R) Gold 6330N CPU @2.20GHz for experimental validation.

3.2.3. Human operators and HMI

Human operators interact with remote equipment and cloud services via dedicated HMIs. In the Avatar framework, a head-mounted display (HMD) is the primary interface, integrating hardware and software to deliver real-time panoramic visuals and sensor data from the teleoperator. This bidirectional channel simultaneously captures operator commands for cloud transmission. Developed using Unreal Engine, the software layer projects stitched panoramic streams onto a virtual spherical surface. Operators are centrally positioned within this sphere, with head-tracking enabling dynamic viewpoint control, which is shown in Fig. 5. Crucially, the HMD's virtual viewpoint maintains precise spatial correspondence with the physical remote environment, ensuring immersive fidelity.

Our test platform features an Intel Core i7_4790 CPU and NVIDIA GTX 2080 GPU, combined with VR peripherals, enabling real-time



Fig. 3. Fisheye frames and the stitched panoramic frame.

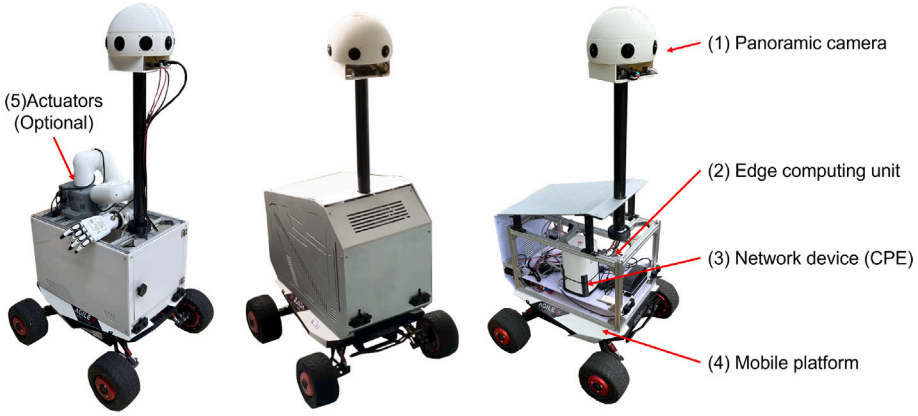


Fig. 4. Prototypes of Avatar teleoperators and core components.



Fig. 5. Human operators and perspective in the Avatar system.

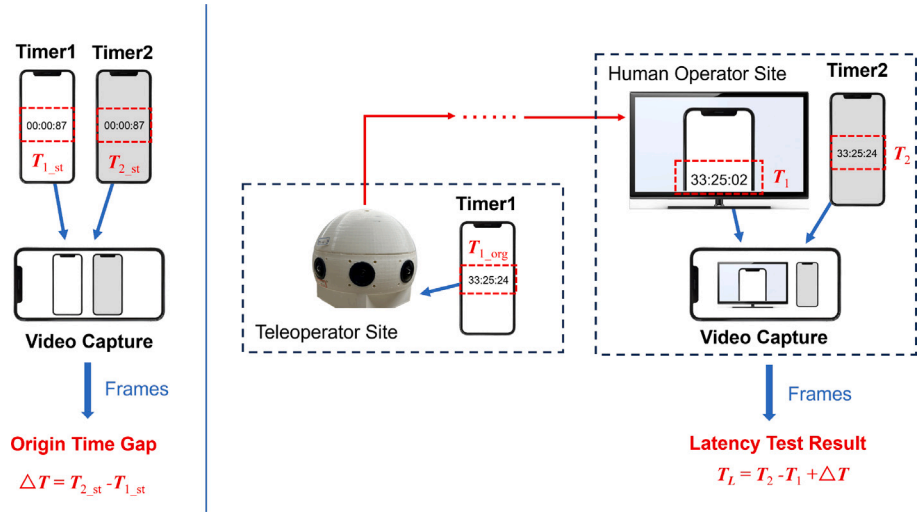


Fig. 6. Test method of capture-to-display latency.

panoramic decoding and 360° immersion. Prototypes evaluations utilized three HMDs: HTC VIVE PRO2 (60 Hz), Pico 4 (60 Hz), and Apple VisionPro (90 Hz), yielding pose latencies of 16.7 ms, 16.7 ms, and 11.1 ms respectively. All values remain below the 17 ms cybersickness mitigation threshold [11].

4. Experiments and results

We will introduce three experimental methods for evaluating the performance of the Avatar system in detail and analyze the results. We will describe optimizing a simple uplink measurement and analyze the system's uplink latency in section. 4.1. Section 4.2 describes a standard test method of response robots on the Avatar. Section 4.3 will focus on teleoperation on dynamic scenes experiments.

4.1. Capture-to-display (uplink) latency

4.1.1. Apparatus and test method of uplink latency

Feldstein [57] proposed a method to measure latency in a teleoperation system. However, their method requires the teleoperator to stay at the same location as the VR kit, which is unsuitable for spatially separated testing. Wei [58] proposed a simple method to test capture-to-display latency in visual systems using two cellphones as timers, allowing spatially separated testing. Similarly, Gruen [59] employed a numerical testing method using self-built timers. We also adopt a similar approach to test the latency based on Wei's method, as shown in Fig. 6.

Our method only needs three cell phones to complete the latency test without changing system devices. A mobile phone capable of capturing video at 240 Hz is used as a capture camera to record video of changing timestamps $\boldsymbol{T}_1$ and $\boldsymbol{T}_2$, with the difference between these timestamps defined as the latency test result at that moment. Two other mobile phones with a 120 Hz LCD refresh rate are used as timers, starting simultaneously before the test. The initial time difference of the timers is recorded in the 240 Hz video clip before and after the latency test to estimate $\triangle \boldsymbol{T}$ and any time drift on the timers. One timer is placed in front of the camera rig on the teleoperator, and its timestamp is captured by the Avatar's vision system and displayed accordingly. Another timer is positioned in front of the display. We use two types of displays in our test: a 240 Hz LCD, and a 60 Hz LCD synchronized with a 60 Hz VR HMD.

According to Wei [58], the timestamps method faces the issue of multiple overlapping timestamps due to the limited response time of the display, which is also influenced by the camera exposure time and the refresh rate of the LCD. To verify the accuracy of our latency test, it is essential to discuss these effects. The analysis of the latencies estimated during the experiment is shown in Fig. 7. The blue dots on the line represent the refresh times of specific components, while the red dots represent the frames and corresponding refresh moments selected for a single calculation. In addition to the latency caused by system processing, the refresh delays in each component contribute to the overall capture-to-display latency, satisfying Equation. (1).

$$\boldsymbol{T}_L = \boldsymbol{T}_{t1} + \boldsymbol{T}_s + \boldsymbol{T}_p + \boldsymbol{T}_d - \boldsymbol{T}_{t2}. \tag{1}$$

$\boldsymbol{T}_{t1}$ is the time gap between the refreshing moment of Timer 1 and the Avatar vision capture. $\boldsymbol{T}_s$ represents the time consumption of a panoramic frame, which includes multi-camera time alignment and stitching. $\boldsymbol{T}_p$ denotes the system processing time, encompassing network transport, decoding, and rendering. $\boldsymbol{T}_d$ is the time gap between the refresh moment of the display and the video captured. Finally, $\boldsymbol{T}_{t2}$ is the time gap between the refresh moment of Timer 2 and the video captured. All of these components fluctuate over time.

In teleoperation or telepresence, the video capture rate represents the time gap of human precepting, which should be continuous in a real application. According to Equation. (1), it does not affect the results of test latency $\boldsymbol{T}_L$, but we found that a higher refresh rate on video capture reduces the timestamps overlapped frames.

In the real world, events occur continuously, implying that:

$$\boldsymbol{T}_{t1} = \boldsymbol{T}_{t2} = 0, \boldsymbol{T}_{real} = \boldsymbol{T}_s + \boldsymbol{T}_d + \boldsymbol{T}_p. \tag{2}$$

According to Equation. (1), if the distribution of $\boldsymbol{T}_{t1}$ and $\boldsymbol{T}_{t2}$ are the same, the average estimation of $\boldsymbol{T}_L$ and $\boldsymbol{T}_{real}$ will be equal, which indicates that we should use the same refresh rate for both timers. A timer with a higher refresh rate can also reduce the variance in the test results and mitigate the issue of overlapping timestamps.

In our test, the capture camera records a video clip lasting 120 s to record the changing timestamps $\boldsymbol{T}_1$ and $\boldsymbol{T}_2$. The video clips are separated into frames using the FFmpeg tool, resulting in 240 frames for each second of video. A Python script is employed to randomly select 200 frames from all the separated frames of the video. The latencies and the original time gaps are estimated by calculating the differences between the timestamps of 100 frames, which are selected from the 200 randomly chosen frames that do not have overlapping timestamps. If insufficient clear frames exist, additional frames are randomly picked until 100 latencies are recorded.

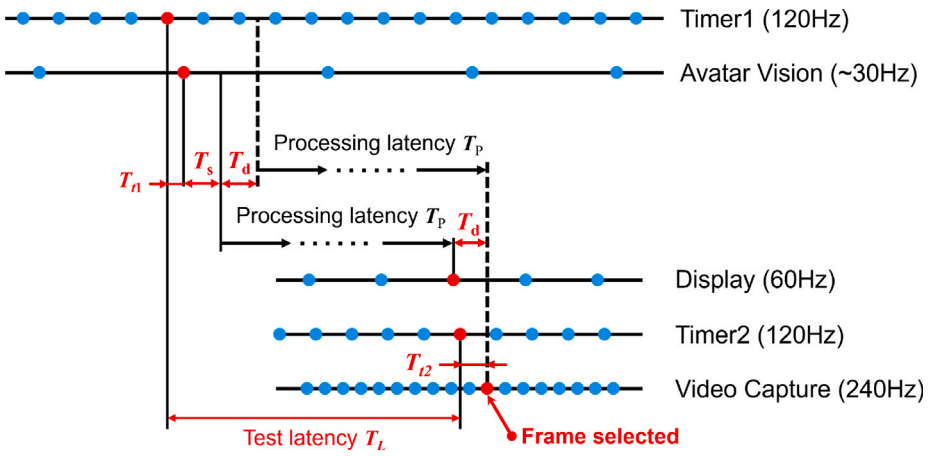


Fig. 7. Latency components in test method.

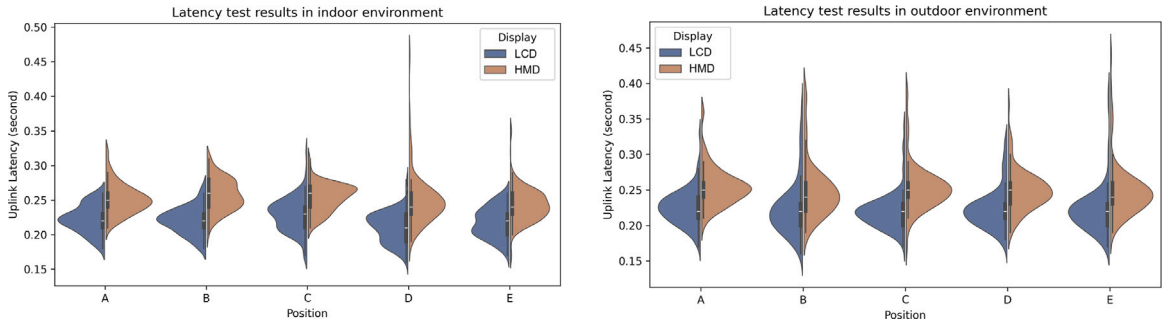


Fig. 8. Violin plots of uplink latencies test results.

Table 1
Latency test results details (seconds).

Position	Point	Display	Maximum	Minimum	Average	STDEV
indoor	A	LCD	0.26	0.18	0.222	0.020
		HMD	0.32	0.21	0.251	0.021
	B	LCD	0.27	0.18	0.220	0.019
		HMD	0.31	0.20	0.257	0.023
	C	LCD	0.32	0.17	0.229	0.024
		HMD	0.31	0.21	0.254	0.020
	D	LCD	0.28	0.16	0.210	0.021
		HMD	0.46	0.19	0.249	0.036
	E	LCD	0.26	0.17	0.217	0.021
		HMD	0.35	0.17	0.242	0.023
outdoor	A	LCD	0.33	0.17	0.225	0.024
		HMD	0.36	0.20	0.254	0.026
	B	LCD	0.37	0.16	0.223	0.038
		HMD	0.39	0.19	0.250	0.040
	C	LCD	0.34	0.17	0.220	0.025
		HMD	0.39	0.17	0.253	0.032
	D	LCD	0.32	0.17	0.223	0.028
		HMD	0.37	0.19	0.249	0.029
	E	LCD	0.39	0.17	0.221	0.032
		HMD	0.44	0.19	0.249	0.038

4.1.2. Latency test results

Latency tests were conducted at ten checkpoints along two teleoperation test trajectories, as illustrated in Fig. 14. Detailed information about latency results can be found in Table 1. The average latencies recorded indoors are 220 ms for the LCD and 251 ms for HMD, while the latencies outdoors are 223 ms for the LCD and 251 ms for the HMD. Violin plots for the latency tests are presented in Fig. 8.

The average latency values do not show significant differences between indoor and outdoor environments; however, the LCD methods are approximately 30 ms lower than those using the HMD. This discrepancy is attributed to the buffer time required for the HMD pose solution. The indoor data is more centralized and exhibits fewer outliers with larger delays, indicating that the network connection is more stable indoors. In contrast, the violin plots for the outdoor data show greater deviations from the mean in the test results, suggesting that the system experiences less stability in terms of delay when operating outdoors. This instability results from network transmission fluctuations and unsynchronized refresh moments among system components.

4.2. Standard test methods of response robot on the Avatar system

Standard test method of response robots [60] are evaluation experiments to measure capabilities for robotic systems proposed by the American National Institute of Standards and Technology (NIST). Compared with complicated tasks, a standard test method can decouple a specific capability from the entire system, providing a more persuasive evaluation result on a specific function. Previous works on teleoperation systems also choose such methods or combinations to evaluate their system performance [7,61].

We selected "visual search in near-field"[61] as evaluation. The purpose of this experiment is to measure the capability of a teleoperator to detect and identify objects within a near-field range and quantitatively evaluate the camera pointing capabilities of the teleoperator.

4.2.1. Apparatus of robots, optronics, and control

Experiment compared the HMI design of Avatar with two mainstream visual HMI solutions currently in use. One is the PTU equipped with a monocular camera and used with LCD, which is the most widely adopted solution in industry and various commercial products, commonly seen in applications such as security surveillance, inspection or rescue robot teleoperation. The other is the PTU equipped with a

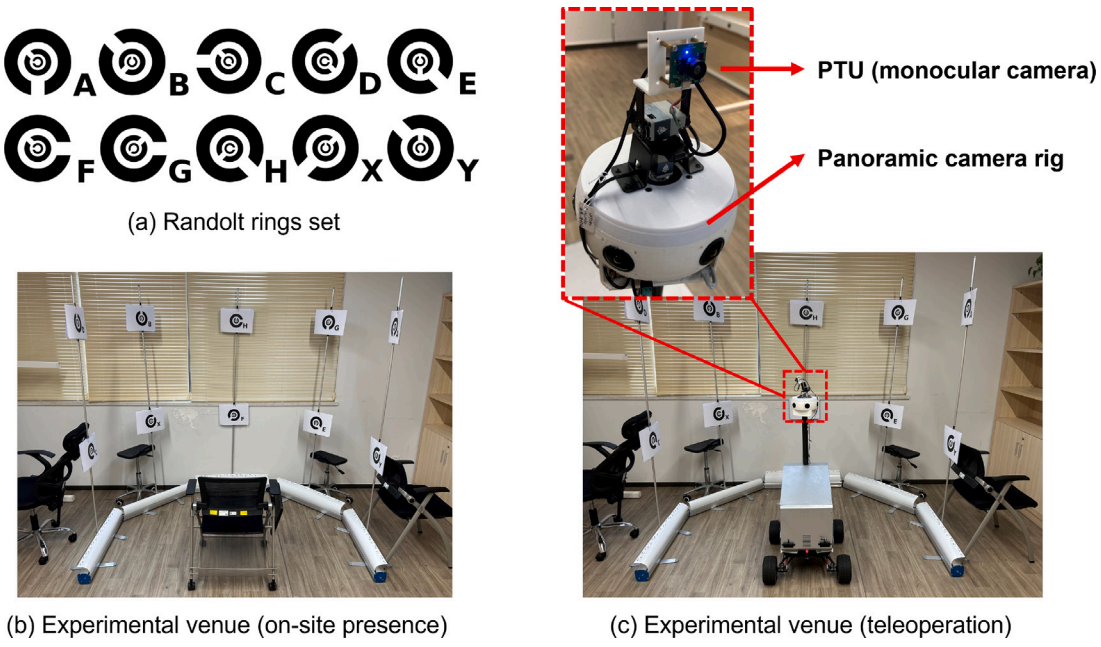


Fig. 9. Apparatus of the near-field visual search.

monocular camera and used with a VR HMD, a solution that is now employed by many immersive teleoperation robot systems, the PTU equipped with a stereo camera can be considered a variant of this approach [32]. This section used an Avatar with a panoramic camera rig and a PTU monocular camera as the teleoperator, as shown in Fig. 9(c). The experiment evaluated three human–machine interface visualization methods: LCD with the PTU, HMD with the PTU, and HMD with a panoramic camera rig. Apparatus of each method are shown in Table 2. The origin uplink latency was about 100 ms for the PTU monocular camera using LCD and 250 ms for the panoramic camera using HMD, tested by the method proposed in Section 4.1. We add uplink latency to the monocular camera to 220 ms for LCD and 250 ms for HMD to synchronize latency for different visualizations. The downlink latency of the PTU camera is about 100 ms, tested by observing the time gap between switching the HMD and the movement of PTU.

The PTU can rotate at an angle in the horizontal direction (yaw angle, assuming the center position is 0°) from −130° to 130° and in the vertical direction (pitch angle) from −74° to 66°. This PTU can ensure FOV coverage larger than 180° in the front direction without rotating the vehicle. The monocular camera on the PTU has a resolution of 1920 × 1080 pixels and a refresh rate of 30 Hz; FOV is 78° horizontal and 47° vertical. The panoramic camera rig is the same as section. 3.1. The height of the monocular camera is 128cm, and the panoramic camera rig is 111cm, similar to a person's eye height when sitting on a chair. We aim to provide similar visual experiences for the participants in different experiments.

The control method of human operators' perspective differs for three visualizations. In the LCD with PTU, the operator's perspective is synthesized with the monocular camera on an LCD, using the keyboard to control the PTU pit and yaw. In the HMD with PTU, the operator's perspective is synthesized with the monocular camera inside the HMD, using head motion captured by the HMD sensors to control the PTU pit and yaw. In the HMD with a panoramic camera rig, the operator's perspective is rendered according to the head motion captured by the HMD sensors, the panoramic camera is fixed.

4.2.2. Experimental design

The test venue consists of ten images. Each image contains a concentric set of three Landolt ring symbols with a randomly oriented direction and a capital English letter, as shown in Fig. 9(a) and (b). The participant must tell the Randolt ring's direction when he/she hears the letter and size (small, middle, or large) from the test host as soon as possible. Each test round contains 30 trials to cover all rings in random order. If the participant correctly answers the direction of the Landolt ring, the next question will be asked immediately. If the answer is incorrect, the test host will remind the participant to confirm again until the answer is correct, and then the next question will be asked. Each participant must complete four test rounds, which contains $\boldsymbol{S}_0$, $\boldsymbol{S}_1$, $\boldsymbol{S}_2$, $\boldsymbol{S}_3$ in Table 2. Firstly on-site presence $\boldsymbol{S}_0$, then using three visualizations $\boldsymbol{S}_1$, $\boldsymbol{S}_2$, $\boldsymbol{S}_3$ in random order. The venue is not observable directly in non-on-site presence tests. Task completion time (TCT) and the number of incorrect direction answers are used as objective indicators. When all tests are finished, the participant is asked to finish a questionnaire to rank the three human–machine interaction methods according to their subjective "preference" and "efficiency" and evaluate the advantages and disadvantages of these three methods subjectively. At the same time, we collected information on whether the participants had VR usage experience and teleoperation experience for analysis.

Volunteers were recruited through online public announcements and poster postings. Each volunteer participated in a 20 to 30-minute test and received a labor remuneration of 30 RMB. We finally invited 28 volunteers to join our primary test to evaluate three human–machine interfaces under similar uplink latency. Among the 28 participants, 22 are male, and six are female. Among them, 23 have VR device experience, and 18 have teleoperation experience. Each participant was informed that all data would be analyzed anonymously, and an experimental consent form was signed with a complete understanding of the experimental content and data analysis. Before each experiment round, participants were given sufficient time to familiarize themselves with each visualization and operation method. Between formal tests, the participant takes a break of 2 min. Simulated tests and rounds of formal tests use different Landolt ring sets to avoid remembering.

To further verify the superiority of panoramic vision, we randomly chose 10 of 28 volunteers to join test $\boldsymbol{S}'_1$, $\boldsymbol{S}'_2$ to see whether panoramic vision works better than traditional PTU with a lower uplink latency of around 100 ms that can be achieved by most of existing teleoperation systems.

Table 2
Apparatus in "visual search in near-field" experiment.

Experiment	Display method	Uplink latency (ms)	Control method
S_0	On-site perception	0	Head movement
S_1	LCD (240 Hz)	220	PTU controlled by keyboard
S_2	HMD	250	PTU synchronized by HMD pose
S_3	HMD	250	Camera fix, perspective switch by HMD pose
S'_1	LCD (240 Hz)	100	PTU controlled by keyboard
S'_2	HMD	130	PTU synchronized by HMD pose

4.2.3. Experimental results and analysis

Since there are individual differences in near-field search capabilities, quantitative indicators are mainly compared among the four experimental results of each individual to evaluate the efficiency of the three different visualization methods fairly. The ratio of the time taken to complete the task using the three different human–machine interfaces to the baseline is used as an indicator reflecting the efficiency, and the comparison of the number of errors with the baseline is used as an indicator of concentration. The results in the near-field task are shown in Table 3 and Fig. 12. Here, baseline refers to the individual's TCT and errors in experiment S_0.

The lower the TCT ratio, the shorter the time the human–machine interface takes to complete the task, which means the higher the efficiency of near-field visual perception. The VR HMD and Avatar panoramic camera method used in Experiment S_3 showed the highest perception efficiency in the experiment. The TCT was only 1.124 times that of on-site perception, which was significantly higher than the traditional schemes of using PTU in combination with a LCD (1.747) or VR HMD (2.160). Seven participants (25% of the total) had a ratio of less than 1.05 in Experiment S_3. They achieved the same efficiency as on-site perception.

A study conducted in a virtual environment [8] indicates that using head-controlled vision aligns more with human perception habits. When deployed in the real world, if the scheme of S_2 is adopted, it will face the problem of "HMD pose latency", which significantly reduces the perception efficiency and may even cause motion sickness. This experimental evaluation performed worse than the traditional human–machine interface in S_1. However, Experiment S_3, which uses the HMD for head-controlled visual direction but adopts the Avatar panoramic vision scheme, reduces the HMD pose latency to the local computing time consumption, effectively ensuring the perception efficiency. This result also proves the remarkable superiority of the panoramic camera deployment scheme in near-field visual search capabilities.

In supplementary experiments S'_1 and S'_2, we restored the uplink latency of the PTU camera to its initial value. At this time, the uplink latency of S'_1 decreased to 100 ms, and that of S'_2 was 130 ms. These are the uplink latencies that most teleoperation systems can achieve when using the current traditional PTU camera scheme. Ten participants were randomly selected from 28 to conduct experiments S'_1 and S'_2, whose results are shown in Fig. 10 and Table 3.

Comparing the results of S_1 and S'_1, as well as S_2 and S'_2, under the premise that the downlink latency is approximately 100 ms, reducing the uplink latency from 200 ms to 100 ms has almost no impact on the near-field visual search efficiency of the deployment scheme of the LCD combined with the PTU camera. The slight increase in the average magnification from 1.747 to 1.809 can be considered within the normal fluctuation range. However, the improvement effect is relatively noticeable for the deployment scheme of using HMD in combination with the PTU camera. Besides the average magnification decreasing from 2.16 to 1.966, the magnification's maximum value, median, and variance decreased significantly. This experimental result proves that the human–machine interface using HMD in combination with the PTU camera is more sensitive to uplink latency. However, under the condition of an end-to-end latency of approximately 200 ms, the near-field visual search efficiency of these two human–machine interfaces using the PTU camera is still far inferior to the deployment scheme of using HMD in combination with the panoramic camera, further demonstrating the superiority of the Avatar visual scheme in near-field visual search. In addition, since the panoramic camera scheme does not involve camera rotation when observing a static near-field environment, it is not affected by the magnitude and fluctuation of the latency, which is another advantage.

Table 3
Details of TCT ratios of visual search in near-field search test.

Experiment	S_1	S_2	S_3	S'_1	S'_2
Average	1.747	2.160	**1.124**	1.809	1.966
STDEV	0.265	0.324	0.110	0.235	0.152
Median	1.719	2.122	**1.110**	1.719	1.941
Maximum	2.610	2.971	**1.398**	2.151	2.257
Minimum	1.285	1.628	**0.956**	1.505	1.778

For incorrect direction answers, only 16 participants made at least one mistake during the whole test, whose results are shown in Fig. 11 and Table 4. This result did not include data on participants who did not make mistakes. Since identifying the direction of the Landolt ring is a relatively clear-cut and straightforward task, on the premise of ensuring that all Landolt rings' directions on the field can be reorganized clearly, we believe that to some extent, the number of incorrect answers is, inversely proportional to the subject's degree of concentration.

The average number of incorrect answers in on-site perception is the highest. We attribute this to the fact that S_0 is the first experiment to start. Although the participants fully understood and familiarized themselves with the process before the formal experiment, the learning effect reduces the possibility of making mistakes in identifying directions in the following experiments. However, we cannot rule out the influence of the participants' insufficient concentration during the first on-site experiment. The three teleoperation experiments were carried out randomly, so the influence of the learning effect can be excluded. The average number of incorrect answers of the two human–machine interfaces S_2 and S_3 that use HMD as the display is slightly lower than that of S_1 with LCD. Although the gap is tiny, we believe this proves that using HMD can give participants a stronger sense of immersion, enabling them to focus more on the task. However, some participants using the HMD human–machine interaction interface made more mistakes. We attribute this phenomenon to the motion sickness caused by the HMD. Fig. 11 shows that, compared with the baseline, the error rates of participants in this task did not change significantly among the three different human-interaction methods. Therefore, the experimental methods did not cause significant cognitive difficulties or obstacles to the participants.

The rankings of the three human–machine interaction methods by the participants in the questionnaire according to "preference" and "efficiency" are shown in Fig. 12. In our experiment, the Avatar human–machine interface using the combination of HMD and panoramic camera ranked first with an absolute advantage in both rankings. The human–machine interface using LCD and PTU combined ranked second, and the human–machine interface using HMD and PTU ranked third. Among them, 22 participants (78.57%) liked this human–machine interface the most, and 23 participants (82.14%) subjectively thought that its perception efficiency was the highest. Interestingly, the remaining five people all subjectively believed that the human–machine

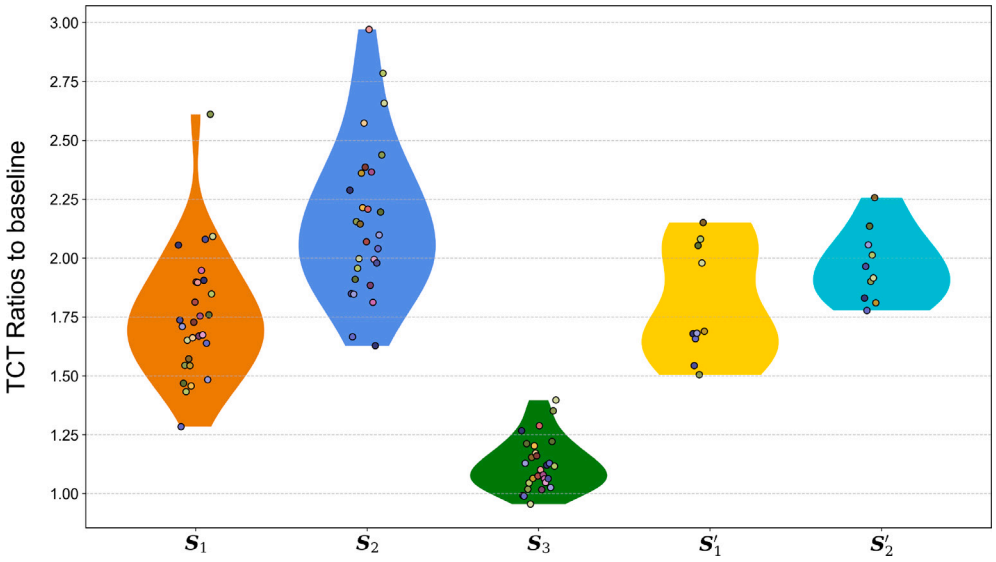


Fig. 10. TCT ratios of visual search in near-field test.

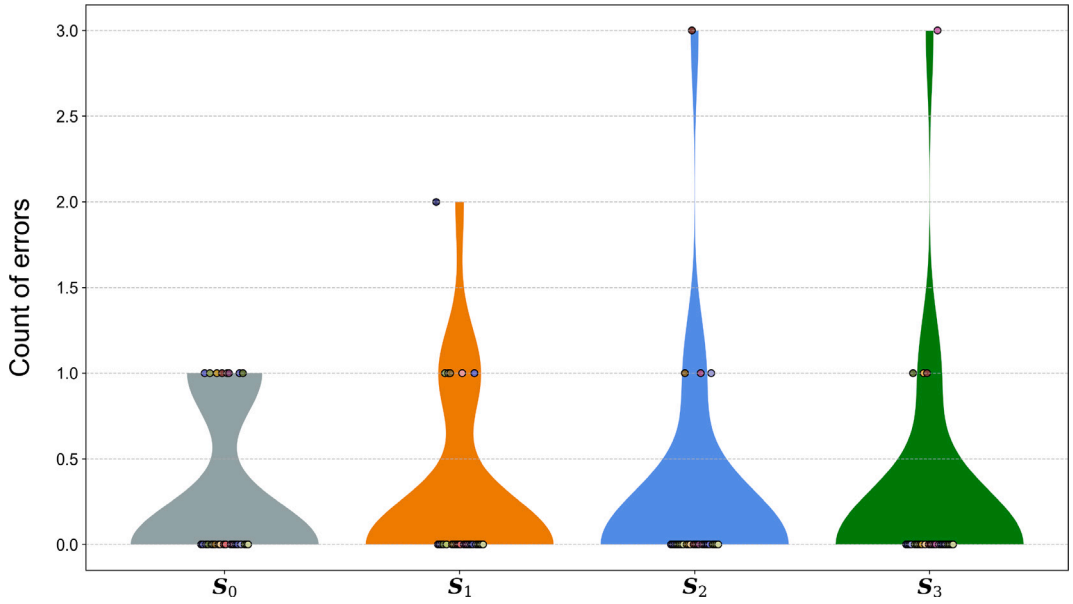


Fig. 11. Count of errors in near-field search test.

Table 4
Details of count of errors in near-field search test.

Experiment	S_0	S_1	S_2	S_3
Average	0.286	0.250	**0.214**	**0.214**
STDEV	0.452	0.509	0.619	0.619
Median	0	0	0	0
Maximum	**1**	2	3	3
Minimum	0	0	0	0

interface of the LCD and PTU combination was more efficient. However, the TCT of the Avatar human–machine interface was the shortest for them.

By checking the subjective evaluations, this wrong subjective perception may stem from two main reasons. One is that the LCD can view multiple patterns simultaneously at certain angles, making participants subjectively think that the efficiency is extremely high. Secondly, there is no need to wear a heavy HMD, which will not cause motion sickness and is relatively comfortable. In addition, other evaluations of LCD with PTU human–machine interface include issues such as steering latency, difficulty in quickly switching the perspective, blurred vision during rotation that makes it impossible to observe, and limited field of view.

For the combination of HMD and PTU, almost every subject mentioned the latency in visual steering and motion sickness, resulting in a poor experience. In addition, the blurring caused by the camera's movement means that participants can only observe after the camera has finished rotating. The coupling with the latency transforms the continuous search-and-observe task into a process of "switch-wait-observe", which significantly reduces the near-field visual search efficiency. Although reducing the latency has improved this problem, eliminating its negative impacts is still impossible. Some users also mentioned that adopting this observation strategy makes head rotation tiring. Moreover, due to the limited resolution of the HMD, some users think its clarity is inferior to that of the LCD.

For the Avatar human–machine interaction method, most users indicated no delay in visual steering with this method. The search efficiency

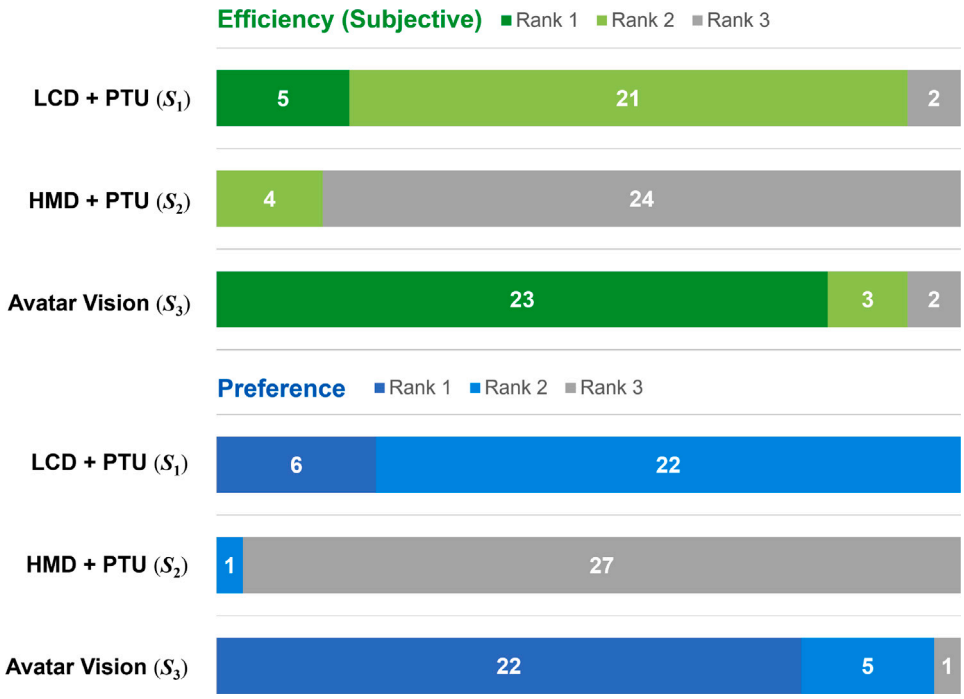


Fig. 12. Rankings of human–machine interactions voted by participants.

is closest to on-site perception; it has a strong sense of immersion and is closest to the on-site perception habit, thus resulting in a low cognitive burden and learning cost. Almost all users believed that there was no apparent physical discomfort caused by motion sickness. Only one person reported mild discomfort, much lower than that caused by the combination of HMD and PTU. However, the main drawbacks of this method are the discontinuous parts in the picture and relatively low clarity, which are caused by the LUTs panoramic stitching method and the resolution limitation of the HMDs. We will improve these issues in our subsequent work.

In summary, in objective analysis and subjective surveys, the Avatar interface is significantly superior to interfaces using LCD or HMD combined with a PTU-camera in terms of near-field visual search efficiency and user preference. Its remarkable advantages include a strong sense of presence, no visual steering delay, and high near-field search efficiency. Its shortcomings are stitching seams and limited resolution.

4.3. Experiment of teleoperation on dynamic scenes

Avatar's panoramic vision has fully demonstrated superiority in the near-field visual search experiment. However, observing static scenes fails to highlight the necessity and advantages of the panoramic vision in terms of low latency. Therefore, we introduce a teleoperation experiment that involves dynamic scenes.

4.3.1. Apparatus of robots, optronics, and control

In these experiments, the Avatar system is deployed on a mobile platform Agile X Scout mini. The control methods are illustrated in Fig. 13. The remoter is the origin controller for the mobile platform, using two joysticks to control linear speed $\boldsymbol{v}$ and angular speed $\boldsymbol{w}$, respectively. A VR grip equipped with a trackpad is employed in the Avatar system to control the teleoperator. The trackpad captures the position of the thumb $(\boldsymbol{X}, \boldsymbol{Y})$ and translates this into linear and angular speeds, as shown in Fig. 13.

For safety considerations, we have established a maximum linear speed of 1.5m/s and a maximum angular speed of 0.5m/s during indoor tests for all control methods. In the outdoor test, the maximum linear speed is increased to 3.0m/s, while the angular speed remains capped at 0.5m/s.

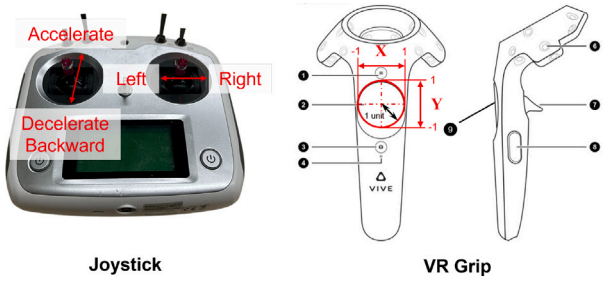


Fig. 13. Control method on the joystick and VR Grip.

Table 5
Apparatus of dynamic teleoperation experiments.

Experiment	Location	Display	Controller
$\boldsymbol{D}_{in_0}$	Indoor	On-site	Joystick
$\boldsymbol{D}_{in_1}$	Indoor	LCD	Joystick
$\boldsymbol{D}_{in_2}$	Indoor	HMD	Joystick
$\boldsymbol{D}_{in_3}$	Indoor	HMD	VR Grip
$\boldsymbol{D}_{out}$	Outdoor	HMD	VR Grip

Given that the PTU-based solution performed inadequately in near-field visual search tasks, we opted to replace the PTU camera with the Avatar panoramic camera in an LCD-based solution to alleviate the challenges of the teleoperation experiment.

4.3.2. Experimental design

In the teleoperation test, participants were instructed to operate the teleoperator using four different apparatuses, each differing in display and control, across five experiments, as outlined in Table 5. The trajectories are shown in Fig. 14. $\boldsymbol{D}_{in_0}$ is a baseline for TCT and the number of collisions, reflecting each subject's teleoperation ability. $\boldsymbol{D}_{in_1}$ and $\boldsymbol{D}_{in_2}$ evaluate the effects of two different display methods. Consider that the remote control is not available at such a physical distance with

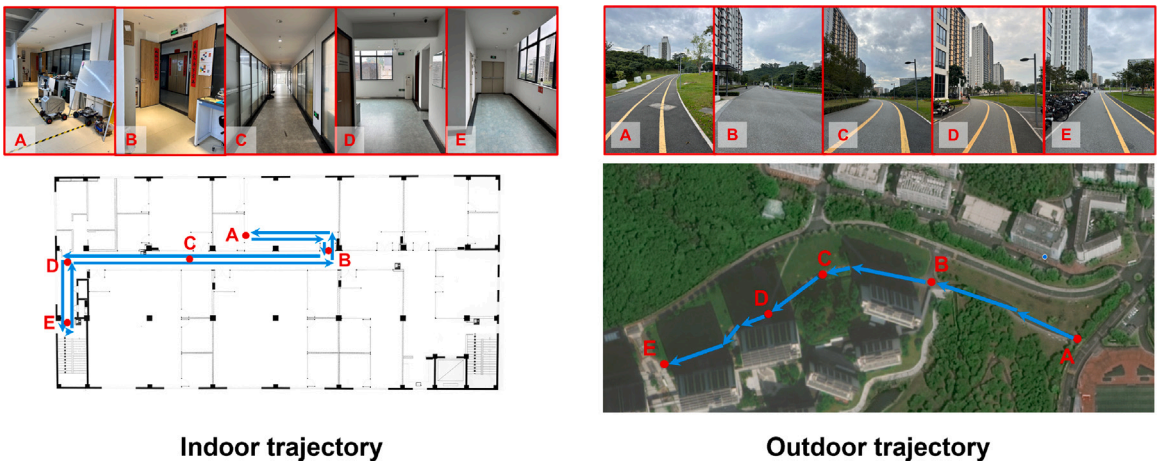


Fig. 14. Trajectories of teleoperation test.

the remoter in $\boldsymbol{D}_{out}$, with the test route being more than 2 kilometers from our laboratory. We design $\boldsymbol{D}_{in_2}$ to assess the influence of the Avatar's VR grip method on teleoperation efficiency when compared with $\boldsymbol{D}_{in_3}$, which can also evaluate the Avatar system's performance in indoor teleoperation tasks. Finally, $\boldsymbol{D}_{out}$ is utilized to assess the Avatar system's performance in outdoor teleoperation scenarios.

We invited 15 volunteers to join in the teleoperation test. The volunteers were recruited from school, comprising 13 males and two females. Each participant had the opportunity to familiarize themselves with the system without any time constraints and could begin the test when they felt ready. Before participating in the experiment, all participants were informed about the data that would be recorded. All participants consented to the use of their data for anonymized analysis. Firstly, in task $\boldsymbol{D}_{in_0}$, participants use the original controller of the teleoperator to complete an on-site operation task following the illustrated route. Subsequently, participants are invited to a room where they cannot see the teleoperator and sequentially complete the teleoperation tasks $\boldsymbol{D}_{in_1}$, $\boldsymbol{D}_{in_2}$, and $\boldsymbol{D}_{in_3}$. Finally, they join the outdoor test $\boldsymbol{D}_{out}$. Participants were asked to complete a questionnaire each time when they finished experiments $\boldsymbol{D}_{in_1}$, $\boldsymbol{D}_{in_3}$ and $\boldsymbol{D}_{out}$.

The evaluation consists of the efficiency and quality of teleoperation and the sense of presence.

To evaluate efficiency and quality, we used TCT ratio to baseline (experiment $\boldsymbol{D}_{in_0}$ in Table 5), the percentage of time spent off-route, and the number of collisions. To avoid the total testing time of a single participant being too long, the outdoor baseline test was conducted only by Tester $No.1$. The outdoor baseline time for the remaining testers was estimated using the ratio of their indoor baseline times to that of Tester $No.1$ in $\boldsymbol{D}_{in_0}$. Indoor tests record the total number of collisions to measure teleoperation quality metrics. In the outdoor test, the teleoperator is driven along a path marked with solid yellow lines on both sides. We assess teleoperation quality by calculating the percentage of time any wheel crosses the yellow lines.

Regarding the sense of presence, there is an ongoing debate about a universally accepted method [62], incorporating both objective and subjective measures is considered more persuasive. We utilize the Slater-Usoh-Steed Questionnaire (SUS) [63], a widely recognized tool for subjectively assessing presence. For objective measures, a multi-conductor physiological recorder made by KINGFAR recorded the electrocardiogram (ECG) and electrodermal activity (EDA) of human operators throughout the experiments. EDA and ECG data are analyzed by Neurokit2[64]. Specifically, we discuss the standard deviation of normal-to-normal intervals (SDNN), the long axis of the ellipse from Poincaré measures (SD2), and the average value of normal-to-normal intervals (Mean NN) from ECG signals, along with the number of skin conductivity reactions per minute (nSCR) from EDA signals. The meanings of each indicator will be explained during the analysis. Due to the significant individual differences in ECG and EDA signals, we recorded 2-minute resting data for each participant as the baseline for the selected indicators. Ratios to baseline will be analyzed.

Table 6
TCT ratios in teleoperation experiments.

Experiment	$\boldsymbol{D}_{in_1}$	$\boldsymbol{D}_{in_2}$	$\boldsymbol{D}_{in_3}$	$\boldsymbol{D}_{out}$
Average	**1.197**	1.431	2.265	1.542
STDEV	0.213	0.289	0.586	0.407
Median	**1.171**	1.466	2.346	1.498
Maximum	**1.593**	2.093	3.272	2.752
Minimum	0.771	**0.765**	0.983	0.893

The ECG and EDA signals, video of the participant's behavior, and video of the teleoperator were recorded simultaneously during all tests. The quantitative indicators for teleoperation efficiency and quality were calculated by reviewing video replays.

4.3.3. Experimental results and analysis

Fig. 15 and Table 6 show the teleoperation efficiency results. Among indoor tests, the HMI combination that achieved the best average teleoperation performance was the LCD based on the Avatar panoramic vision $\boldsymbol{D}_{in_1}$, which took an average of 1.197 times as the baseline. $\boldsymbol{D}_{in_2}$ using HMD based on the Avatar panoramic vision ranked second, with an average time of 1.431 times as baseline. Four participants drove faster in $\boldsymbol{D}_{in_2}$ than $\boldsymbol{D}_{in_1}$. Experiments $\boldsymbol{D}_{in_1}$ and $\boldsymbol{D}_{in_2}$ eliminated the effects of downlink latency and control methods, the LCD performed better than the HMD, which is contrary to the results of the near-field search experiment. We summarize the reasons as follows: Firstly, in teleoperation tasks, it is often only necessary to pay attention to the front, and there is no need to change the viewing direction frequently. Without rotation, the field of view of the LCD can meet the teleoperation needs, offsetting the advantage of the HMD method in near-field search. Secondly, during dynamic teleoperation, the camera shakes and jolts. The conflict between vision and the vestibular sense is more likely to cause motion sickness in the immersive space of the HMD, having a negative impact. This reason is inconsistent with the cause in Experiment $\boldsymbol{S}_2$, which is caused by "HMD pose latency". Compared with Experiment $\boldsymbol{S}_3$, a higher proportion of participants reported mild motion sickness, confirming this inference. Nonetheless, the experiment still demonstrated that Avatar's ultra-low-latency panoramic vision system can meet the requirements of teleoperation. Whether using an LCD or an HMD, all participants completed the teleoperation tests satisfactorily in a relatively narrow and complex indoor trajectory.

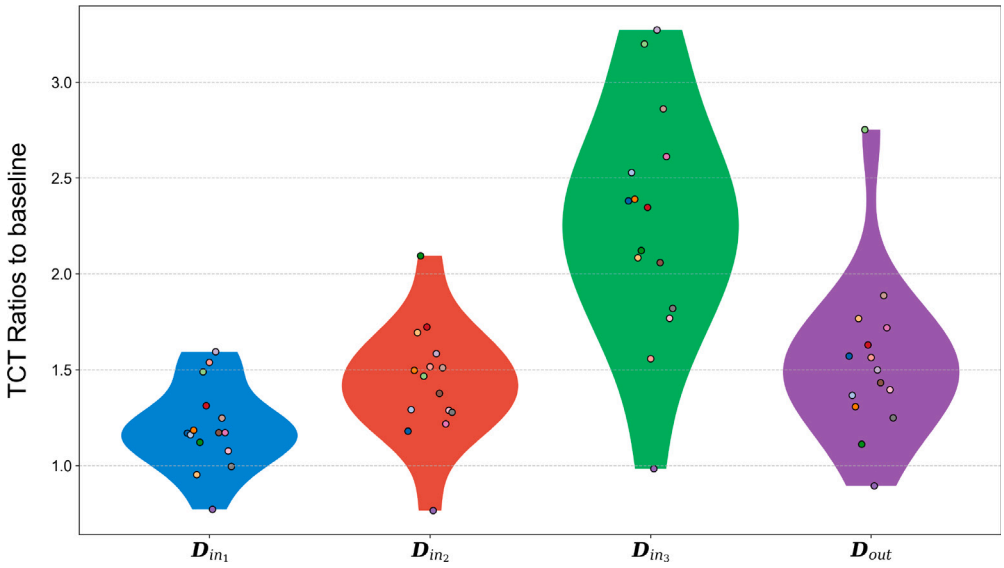


Fig. 15. Teleoperation efficiency indicator: TCT ratios to baseline.

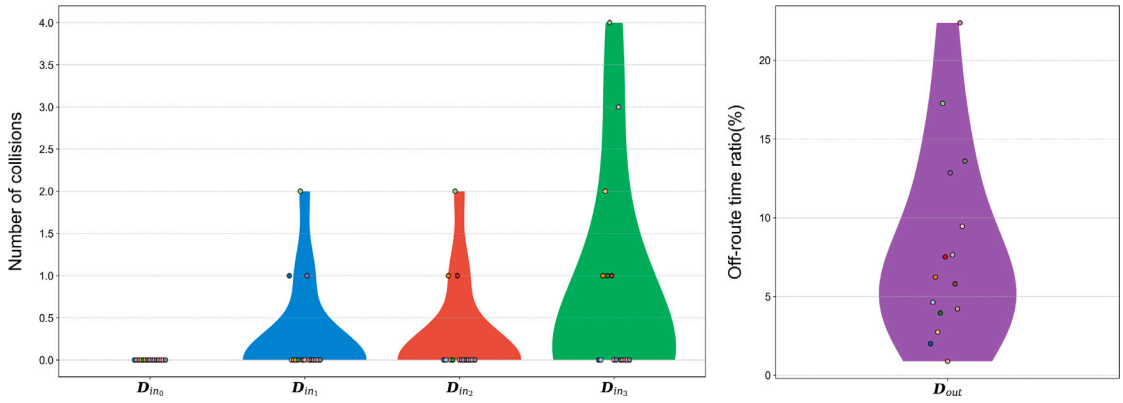


Fig. 16. Teleoperation quality indicators: collisions and off-route time.

Table 7
Details of teleoperation quality indicators.

Experiment	D_{in_0}	D_{in_1}	D_{in_2}	D_{in_3}	D_{out}
	Collisions ↓				*Off-route Time* ↓
Average	0	0.267	0.267	0.800	8.077%
STDEV	0	0.573	0.573	1.222	5.856%
Median	0	0	0	0	6.230%
Maximum	0	2	2	4	22.380%
Minimum	0	0	0	0	0.880%

Due to the physical-distance constraints present in the outdoor experiments, we were unable to use the mobile platform's native controller. For reasons of system compatibility and minimizing user learning cost, we therefore adopted the widely available VR grip as the control device. Experiments D_{in_3} and D_{in_2} were conducted to demonstrate that the VR grip control method does have certain limitations. Results showed that the VR grip control method has significantly reduced the teleoperation efficiency. However, these limitations are attributable to the control method itself rather than to the visual perception framework designed for the Avatar system. Teleoperation efficiency using Avatar system outdoors in D_{out} was superior to the indoor one D_{in_3}, approaching the efficiency in the D_{in_2} experiment. We attribute the differences between indoor and outdoor testing primarily to the narrower and more complex indoor route, which contains obstacles and interferences. All participants demonstrated greater coherence and fluency during outdoor tests. Video observations revealed that the complexity of the indoor route required frequently controlling the vehicle for steering and avoidance maneuvers, which further amplified the performance differences between our designed VR grip and the original remote control.

Generally, the TCT ratios should be greater than 1. However, for Tester *No.*9, the ratios are less than 1 due to his/her physical limitations in walking, which made the on-site operation task more challenging for him/her, resulting in a lower TCT ratio than 1. This result also, to some extent, indicates that the Avatar system can provide convenience for people with disabilities.

For quality indicators, results are shown in Fig. 16 and Table 7. The collision situations in Experiments D_{in_1} and D_{in_2} were basically the same. The vast majority of participants did not experience collisions, which proves the importance of the ultra-low-latency vision system in ensuring teleoperation quality. However, the average number of collisions in D_{in_3} was significantly higher, demonstrating that the VR Grip control method has a negative impact. Since there is no baseline for the outdoor quality indicators of each participant, only a qualitative analysis was carried out for the Dout experimental group. The time of deviation from the boundary of the vast majority of drivers was less than 10%. By watching the video, it was found that drivers could slow down or stop in time and correct the direction in most cases of deviation from the center line. Although the control method introduced some negative impacts, it also shows that the ultra-low-latency panoramic

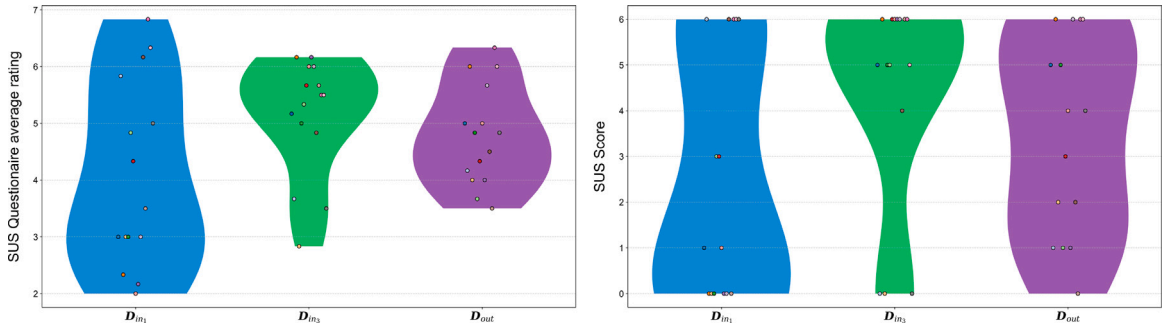


Fig. 17. SUS questionnaire ratings and SUS scores.

Table 8
Details of SUS questionnaire ratings and SUS scores.

Experiment	D_{in_1}	D_{in_3}	D_{out}
SUS Questionnaire Ratings ↑			
Average	4.089	**5.133**	4.789
STDEV	1.583	0.991	0.857
Median	3.500	**5.500**	4.833
Maximum	**6.833**	6.167	6.333
Minimum	2.000	2.833	**3.500**
SUS scores ↑			
Average	2.533	**4.400**	3.467
STDEV	2.630	2.274	2.093
Median	1	**5**	4
Maximum	6	6	6
Minimum	0	0	0

Table 9
Details of ECG and EDA results.

Experiment	D_{in_0}	D_{in_1}	D_{in_2}	D_{in_3}	D_{out}
nSCR ↑					
Average	1.425	0.957	1.267	2.290	**2.659**
STDEV	0.842	0.537	0.770	1.197	2.189
Median	1.300	0.909	1.000	**2.429**	2.200
Maximum	3.080	2.429	2.600	4.667	**9.920**
Minimum	0.400	0.222	0.250	0.556	**0.833**
Mean NN ↓					
Average	0.839	1.054	1.010	1.001	**0.994**
STDEV	0.056	0.051	0.043	0.106	0.098
Median	0.826	1.052	1.004	0.988	**0.987**
Maximum	0.942	1.152	**1.087**	1.200	1.187
Minimum	0.721	0.986	0.928	**0.776**	0.791
SD2 ↓					
Average	0.944	0.869	**0.738**	0.769	0.822
STDEV	0.561	0.358	0.224	0.277	0.463
Median	0.759	0.773	0.743	0.729	**0.667**
Maximum	2.356	1.598	**1.165**	1.393	2.166
Minimum	0.245	0.400	0.447	0.435	**0.348**
SDNN ↓					
Average	1.112	0.936	**0.760**	0.825	0.949
STDEV	0.823	0.419	0.215	0.295	0.804
Median	0.893	0.810	**0.760**	0.765	0.666
Maximum	3.393	2.115	**1.165**	1.441	3.695
Minimum	0.249	0.428	0.501	0.451	**0.365**

vision system played an important role. Tester No. 6 experienced more collisions and off-route time due to moderate simulation sickness, while Tester No. 13 reached the highest off-route time due to moving the teleoperator too fast.

For subjective evaluations on immersion and sense of presence, the SUS score is calculated by counting the number of questions that participants rated "5" or higher on six questions, indicating a high sense of presence, as shown in Fig. 17 and Table 8. Experiment D_{in_3} achieved the highest average SUS score. From the data distribution, it can be seen that more participants considered the HMI in this experiment to have the highest sense of immersion. D_{out} ranked second. Through inquiries, the reason was that the latency in the outdoor experiment was not as stable as that in the indoor experiment. The resulting lag and blurred images reduced the sense of immersion. A few participants also reported that teleoperation through familiar or previously visited locations enhanced their senses of presence when viewing the panoramic stream through the HMD. Although the stitching based on the LUTs method caused flaws at the seams of the panoramic stream, Avatar's ultra-low-latency HMD panoramic vision solution eliminated the HMD pose latency, making users' spatial perception more in line with intuitive feedback and memory to such locations, thus achieving a better sense of immersion. This result is consistent with the conclusion mentioned in Fiala's research [9].

Objective evaluation indicators are shown in Fig. 18 and Table 9. nSCR relates to emotional arousal and cognitive processing [65], while Mean NN reflects the emotional arousal level and nervous system activity. Both SDNN and SD2 reflect the sympathetic modulation [66]. The arrow directions beside the names of various indicators in Table 9 represent the data change trends when the sense of immersion is enhanced, emotional arousal increases, and becomes more concentrated. Since in the baseline D_{in_0}, the participants were required to walk following the vehicle, which is different from their physical state of sitting on a chair in the subsequent teleoperation experiments, we only list the indicators of D_{in_0} (gray in color) but do not compare them with those of the formal experiments.

In the order of D_{in_1}, D_{in_2}, D_{in_3}, and D_{out}, the average ratios of Mean NN were consistent in the four experiments, showing a very slight downward trend, indicating that the heart rates of the participants were relatively stable but slightly increased. The average ratio of nSCR showed a significant upward trend, indicating a significant increase in emotional arousal and a higher degree of cognitive resource occupation. The average ratios of the two indicators, SD2 and SDNN, to the baseline, decreased in all four experiments, being the most obvious in D_{in_2} and the second most obvious in D_{in_3}, indicating that the sympathetic nerve activities in these two experiments were significantly enhanced. It can be considered that the participants' emotional arousal increased, and the stress or cognitive burden increased to some extent.

The combination of these indicator changes is sufficient to prove that during the experiments, the participants were mainly in a state of immersive focus accompanied by tension but with relatively low physical activity. The average immersion level in the three HMD experiments was higher than in the LCD experiment. Moreover, due to the operational burden introduced by VR Grip, the participants were more nervous in Experiments D_{in_3} and D_{out}. Due to the longer distance in the outdoor experiment, the need to concentrate for an extended period made the changes in the indicators in D_{out} more significant. This result is highly consistent with the analysis of the teleoperation assessment and the results of the SUS questionnaires.

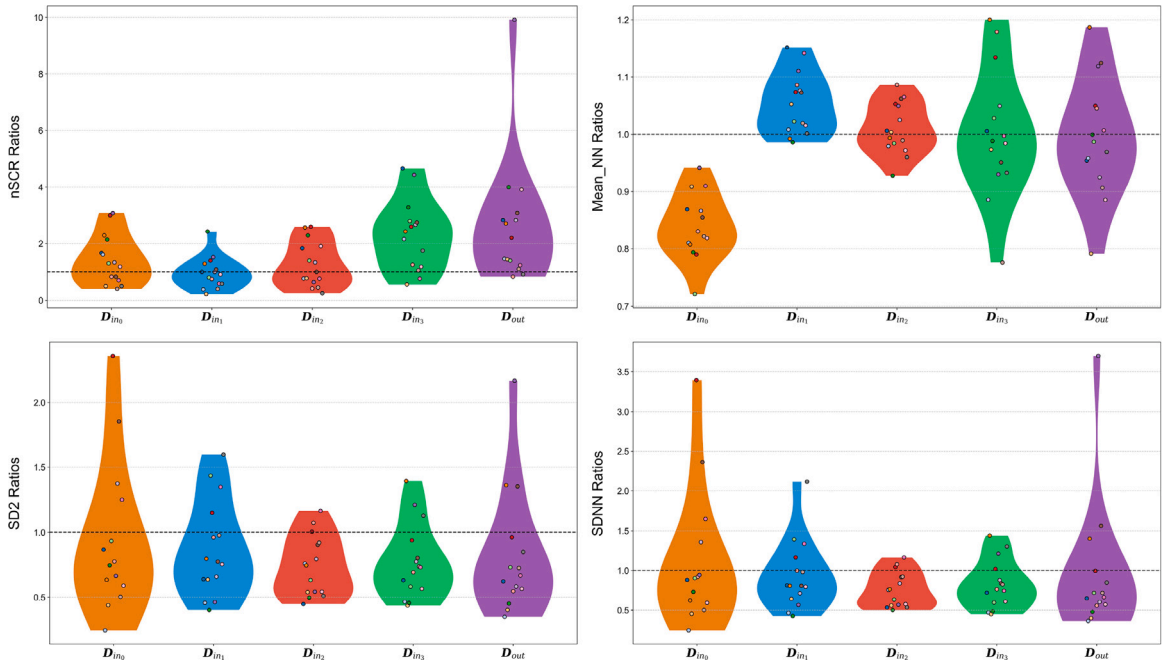


Fig. 18. Physiological indicators in teleoperation tests.

Considering various evaluation results comprehensively, the Avatar system ensured the execution efficiency and quality of the teleoperation task. It also gives operators a strong sense of immersion, enabling them to concentrate on teleoperation. However, the method we designed to control the mobile platform remotely introduced obvious burdens to the participants, causing a significant increase in stress and reducing the execution efficiency and quality of the task. We will address this limitation in our subsequent work.

5. Conclusion

This research proposed the Avatar system, an ultra-low-latency panoramic immersive human–machine interaction platform designed and built for mobile robots. We described the key steps to achieve an ultra-low-latency immersive HMI. In the experiment, the measured panoramic uplink latency of the prototype Avatar system can be as low as 220 ms for LCD and 250 ms for HMD, the lowest reported in the existing formal literature. It can meet most teleoperation tasks' real-time requirements. In the near-field visual search standard test, Avatar demonstrated a perception efficiency close to being on-site, showing significant advantages over the traditional HMI scheme using PTU cameras, highlighting the advantages of the panoramic camera HMI scheme. In the teleoperation experiment, the ultra-low uplink latency enabled the Avatar system to achieve good teleoperation quality and efficiency. Subjective and objective indicators further verified that the system provided a strong sense of immersion and presence. However, due to limitations such as motion sickness caused by the immersive HMI and the cognitive burden introduced by the designed control method, the HMD HMI of the Avatar system is less efficient in teleoperation than using LCD combined with the original joystick controller.

The outstanding performance of the Avatar system, achieved using existing mature hardware, is primarily due to the immersive system design and latency optimizations across multiple components of the uplink process, including both algorithm design and hardware improvements. Additionally, the use of a panoramic camera for video streaming effectively addresses the HMD pose latency issue commonly found in non-panoramic streaming solutions, significantly enhancing user immersion and reducing motion sickness, thereby improving the overall human–machine interaction experience. The Avatar system will inspire exploring the optimal universal HMI system and play a key role in the HCI in emerging technology fields such as the metaverse and embodied intelligence.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhaoyuan Ma reports financial support was provided by Shenzhen Municipal People's Government. This paper relates to a FTC 2025 conference paper "Amplifying robotics capacities with a human touch: an immersive human–machine interaction system". This conference paper, differs substantially in focus and contribution: 1. FTC 2025 Paper: Focuses on system components, expandable functions, deployed mobile robots, qualitative conclusions, and mentions system delay measurements. 2. This Paper: Focuses on evaluating the effectiveness of Avatar HMI in human–computer interaction. It introduces two novel task-specific experiments assessing participant performance. While system latency and composition are briefly covered in both, it provides a detailed analysis of the latency measurement methodology. System composition is described only to the extent necessary for understanding its operation. It is not a primary research focus in the experiments. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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